

DATA STRUCTURES

Infix, Postfix, Prefix.

$A+B$, $AB+$, $+AB$

=> Stack

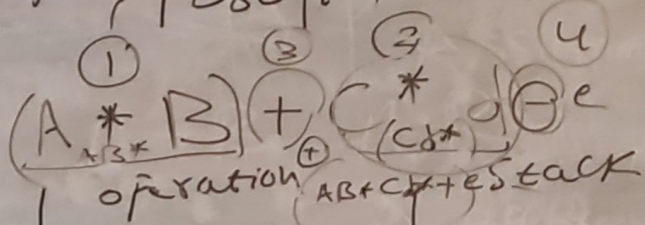
$A+B * C$ $\xrightarrow{\text{Postfix}}$ $A \oplus B C *$ $\Rightarrow$ $AB C * +$

Rules:

Symbol	operation	Stack	output
A	—	—	A
+	push	+	A
B	—	+	AB
*	push	+ *	AB
C	—	+ * C	AB C
End	pop	+ * $\leftarrow$	AB C * =

+!

Infix, Postfix, Prefix.



sy.

A
*
B
+
(
*
d
-
e
int

P

P

P

P

P

P

*

*

+

+

+

+

$\Rightarrow$ $AB * Cd * e - +$
output

A

A

AB

AB*

AB*C

AB*d

AB*d

AB*d*

AB*d*-

AB*d*-e

Rules

Rules:

20/2

sy

op

Sta

del

-

P

-

AB*d

DATA STRUCTURE

Infix, Postfix, Prefix.

$$\begin{array}{cccc} \textcircled{1} & \textcircled{3} & \textcircled{2} & \textcircled{4} \\ (A * B) + C * d - e \\ \text{1 operation} & \text{AB+CD+e stack} \end{array}$$

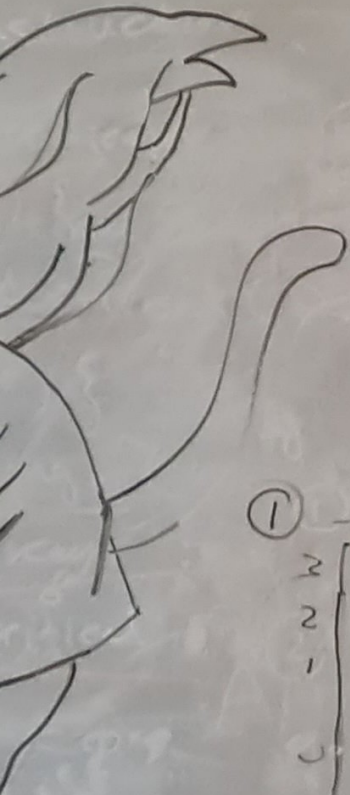
$$\Rightarrow AB * Cd * \text{output}$$

$$\textcircled{1} (AB *) (+) (Cd *) (-) e$$

$$\textcircled{2} (AB *) (Cd *) + (-) e$$

$$AB * Cd * + e -$$

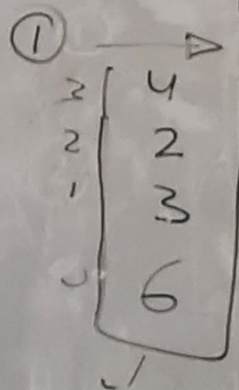
DATA STRUCTURES



$$(5+3) * (4+2) = 8 * 8 = 64$$

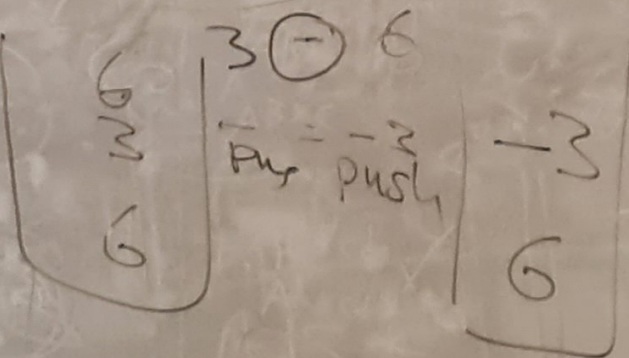
Post

6 3 2 4 + - *



2 (+)
First elem
pop()

②
4 = 6
Second elem



$$6 * -3 = -18$$

Stack

FILO
LIFO

$$(A) + (15)$$

DATA STRUCTURE

$$(5+3) * (4+2) = 8 * 8 = 64$$

$$(a+b) / (c * d)$$

$$a = 10$$

$$b = 3$$

$$c = 1$$

①

2
2
1

3 2 4 + - *

2 (+)
First element
PopC)

4 = 6
Second element

②

6
3
6

3 (-)
Push

-3
6

$$6 * -3$$

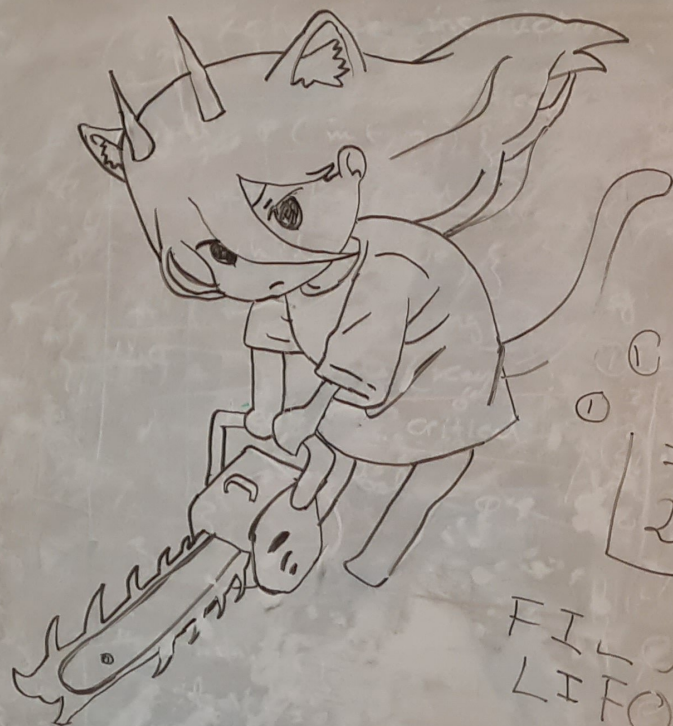
$$= -18$$

Stack

FIL
LIFO

$$(A) + (15)$$

DATA STRUCTURES

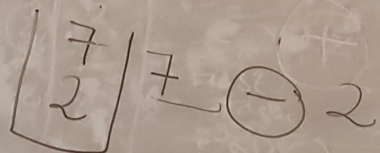


evaluation
prefix

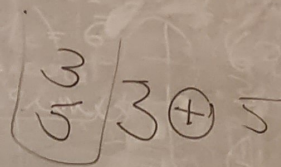
$$+ 3 - 7 2 = ?$$

① Reverse - 2 7 - 3 +

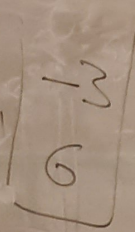
①



FIL
LIFO



3 - 6
push



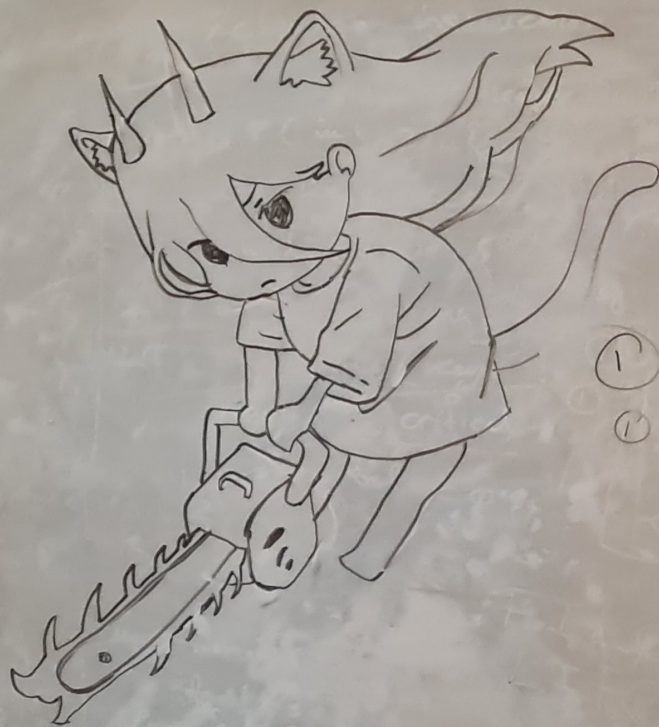
6 * -3
= -18
Stack

$$① (a+b) / (c * d)$$

se a = 10 c = 1
R = 5

Rule 2

DATA STRUCTURES



$+-abc = ?$

if $a=5, b=4, c=3$ se

① $cba \ominus +$

①

5
4
3

5 - 4

②

1
3

③

4

1

+

3 =

① $(a+b) / (c * d)$

$a=10$

$c=1$

$b=5$

3 - 6
5
push

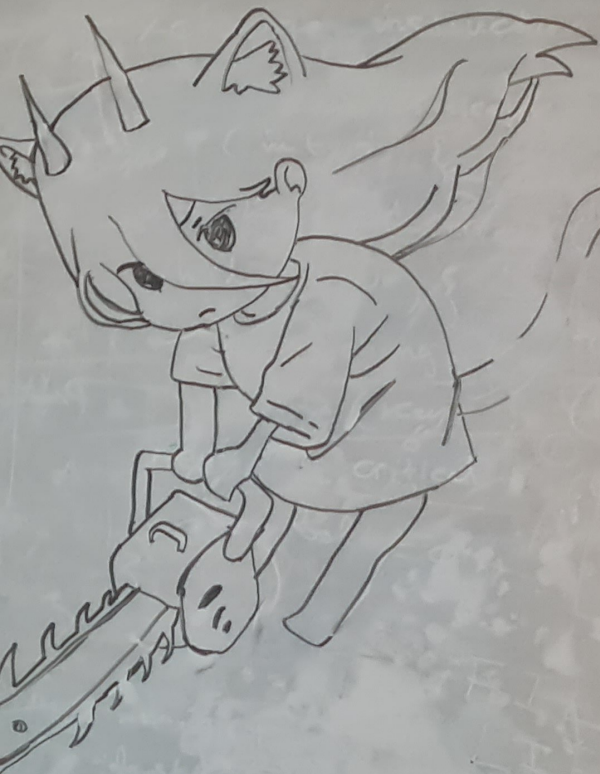
8

-3
6

6 * -3

= -18

step -18



postfix

$$(A+B) * C \Rightarrow (AB+) * C$$

eg.

(
A
+
B
)
*
C
end

operat.
push

—

push

—

pop

push
pop

stack

(

(

(+

(+

+
*

*

output

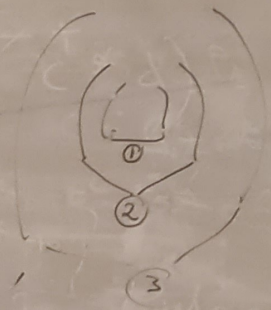
A

A

AB

AB +

AB + C *
= A + C * x



DATA STRUCTURES

$$① A - (B^{\wedge} D)^{\wedge} C$$

$$② A / B^{\wedge} C * D^{\wedge} E$$

$$③ A - B^{\wedge} C + D$$